RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Friday, 14 August 2026, 9:07 AM
State	Finished
Completed on	Friday, 14 August 2026, 9:07 AM
Time taken	37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Recursive Binary Search implementing the Divide and Conquer strategy
int binarySearch(int arr[], int low, int high, int target) {
    if (low <= high) {
        int mid = low + (high - low) / 2;

        if (arr[mid] == target) {
            return mid;
        } else if (arr[mid] > target) {
            return binarySearch(arr, low, mid - 1, target);
        } else {
            return binarySearch(arr, mid + 1, high, target);
        }
    }
    return -1;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) {
        return 0;
    }

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    int found = 0;
    int el1 = -1, el2 = -1;

    // Iterate through the array and use recursive D&C binary search for the complemen
    for (int i = 0; i < n; i++) {
        int complement = x - arr[i];
        int idx = binarySearch(arr, i + 1, n - 1, complement);

        if (idx != -1) {
            el1 = arr[i];
            el2 = arr[idx];
            found = 1;
            break;
        }
    }

    if (found) {
        printf("%d\n%d\n", el1, el2);
    } else {
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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